

Linking above- and belowground plant phenology to carbon cycling in peatlands



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Project overview:

This PhD project aims to quantify how belowground plant phenology and root activity regulate carbon cycling in peatlands, and to determine how the timing of root growth and turnover aligns - or becomes decoupled - from aboveground green leaf phenology. By explicitly linking seasonal dynamics of canopy development with belowground production and activity, the project will assess how shifts in plant timing influence ecosystem carbon exchange.

Using a minirhizotron-based approach combined with measurements of aboveground green leaf phenology, CO₂ and CH₄ flux measurements, isotopic tracers, and process-based analyses, the student will directly couple canopy development with belowground root dynamics to investigate how seasonal plant activity drives greenhouse gas exchange across contrasting plant communities. These plot-scale measurements will be integrated with site-level flux observations and remote sensing approaches to upscale phenological controls on peatland carbon dynamics under changing environmental conditions.

Research objectives:

1. **Track above and belowground phenology and root dynamics:**
Install and monitor minirhizotrons to quantify seasonal patterns of root growth, turnover, and activity, and assess their timing relative to aboveground phenology.
2. **Quantify carbon flux partitioning:**
Measure CO₂ and CH₄ fluxes across the growing season and combine isotopic and process-based measurements to distinguish root-derived carbon inputs from peat decomposition.
3. **Assess spatial variability across plant communities:**
Compare root dynamics and carbon cycling across moisture and microtopographic gradients to determine how plant community composition influences ecosystem-scale fluxes.
4. **Integrate spectral and remote sensing approaches:**
Collect frequent field spectroscopy and proximal sensing data to track fine-scale phenology and link ground measurements with broader-scale observations.

Candidate requirements:

- MSc degree in ecology, environmental science, biogeochemistry, soil science, geography, hydrology, or a related field.
- Experience with fieldwork in remote environments.
- Interest in plant–soil interactions, carbon cycling, and peatland ecosystems.
- Experience with greenhouse gas flux measurements, isotopes, or root ecology (an asset but not required).
- Strong quantitative and communication skills.

Training and opportunities:

The student will receive training in aboveground phenology measurements, minirhizotron imaging, carbon flux measurements, isotopic techniques, spectroscopy, and data analysis. The project offers opportunities for interdisciplinary collaboration, international partnerships, and participation in broader peatland carbon research networks.

How to apply:

Please submit the following as a single PDF file to Scott J. Davidson davidson.scott_j@uqam.ca:

- Cover letter outlining research interests and motivation for the position
- Academic CV (including publications, field/lab experience, technical skills)
- Transcripts
- Names and contact information of two referees

Applications will be reviewed on a rolling basis until the position is filled.